**Zen AI Model Family**

# Zen4-Thinking

Fourth-Generation Extended Reasoning Model

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**Abstract**

We present **Zen4-Thinking**, a 72B parameter model capable of extended chain-of-thought reasoning with an explicit thinking token budget of up to 32,768 tokens before producing a final response. Zen4-Thinking is trained through a novel **Budgeted Reflection Optimization** (BRO) framework that teaches the model to allocate thinking tokens efficiently across problem difficulty, avoiding both under-thinking on hard problems and over-thinking on easy ones. The model achieves 93.7% on MATH, 87.5% on AIME 2024, 81.3% on GPQA Diamond, 72.4% on USAMO, and 88.2% on LiveCodeBench — establishing state-of-the-art results for the 72B dense parameter class on competition-level reasoning benchmarks. Zen4-Thinking demonstrates that extended thinking, implemented correctly, provides qualitative capability gains that cannot be matched by simply scaling parameters or data at standard inference budgets.

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# 1 Introduction

Mathematical competition problems and advanced scientific reasoning tasks share a structural property that makes them resistant to standard language model approaches: the correct answer depends not on memorized associations but on a sequence of inferential steps, each of which must be logically valid for the final answer to be correct. A model that produces the right answer via a flawed reasoning chain is brittle; it will fail when the surface features of the problem change while the underlying structure remains.

Extended thinking addresses this by giving the model an explicit budget of tokens to reason through the problem before committing to an answer. The thinking tokens are not shown to the user; they serve as a scratchpad where the model can:

- Explore multiple solution approaches before committing to one.
- Verify intermediate steps by re-deriving them from different angles.
- Identify and correct errors in prior reasoning.
- Enumerate cases systematically rather than guessing at coverage.

Prior work on chain-of-thought prompting [**?**] established that interleaving reasoning with answers improves performance on multi-step tasks. Extended thinking scales this insight: rather than a few hundred tokens of reasoning, Zen4-Thinking supports up to 32,768 thinking tokens. This qualitative increase in reasoning budget enables solution strategies that are simply not expressible in shorter reasoning chains.

## 1.1 Key Contributions

- **Budgeted Reflection Optimization (BRO)**: A new post-training framework that teaches the model to allocate thinking budget proportionally to problem difficulty, measured by the outcome variance across rollouts at a given budget.
- **32K Thinking Token Support**: Native architectural support for thinking tokens up to 32,768 with no penalty to standard generation quality.
- **Thinking-Aware Quantization**: A quantization scheme that preserves higher precision in the layers most heavily used during thinking token generation.
- **State-of-the-Art Competition Math**: 87.5% on AIME 2024, 72.4% on USAMO averaged across 2024–2025 competition sets.

## 1.2 Thinking vs. Standard Mode

Zen4-Thinking operates in two modes:

**Standard mode** The model generates a direct response without thinking tokens, equivalent to Zen4-Pro behavior. This mode is used when thinking is disabled by the caller or when the model determines the query does not require extended reasoning.

**Thinking mode** The model generates a `<think>...</think>` block containing up to 32K tokens of internal reasoning before generating the final response. The thinking block is generated autoregressively and is subject to the same KV-cache and context window constraints as regular tokens.

The model selects between modes based on the input query when `thinking=auto`, or the caller can enforce `thinking=enabled` or `thinking=disabled`.

# 2 Architecture

## 2.1 Model Specifications

| Component | Specification |
|---|---|
| Total Parameters | 72.7B |
| Active Parameters | 72.7B (dense) |
| Architecture | Zen MoDE Hybrid Transformer |
| Attention Mechanism | Hierarchical Hybrid Attention (HHA) |
| Local Window Size | 8,192 tokens |
| Global Anchor Tokens | 1,024 per layer |
| Context Length (total) | 131,072 tokens (128K) |
| Max Thinking Tokens | 32,768 |
| Max Response Tokens | 98,304 (128K minus thinking) |
| Hidden Dimension | 8,192 |
| Intermediate Dimension | 29,568 |
| Attention Heads | 64 |
| Key-Value Heads | 8 (GQA) |
| Layers | 80 |
| Vocabulary Size | 151,936 |
| Thinking Token IDs | Special tokens: `<think>` (ID 151668), `</think>` (ID 151669) |
| Positional Encoding | RoPE ($\theta = 10{,}000{,}000$) |
| Normalization | RMSNorm |
| Activation | SwiGLU |

Table 1: Zen4-Thinking Architecture Specifications

## 2.2 Thinking Token Architecture

Thinking tokens are native to the Zen4-Thinking architecture from the beginning of training. Two special tokens, `<think>` and `</think>`, are added to the vocabulary. During generation with thinking enabled, the model generates `<think>` followed by a variable-length chain of reasoning, then `</think>`, and finally the user-visible response.

Critically, thinking tokens are treated as fully ordinary tokens by the attention mechanism. The HHA local and global attention applies identically to thinking tokens, reasoning tokens, and response tokens. This means:

- Reasoning steps in the thinking block can attend to all prior thinking steps.
- The response generation can attend to the entire thinking block, ensuring the reasoning is fully available during answer synthesis.
- The KV-cache for thinking tokens is held in memory and can be inspected for debugging; the cache can optionally be offloaded if response-only latency is more critical than thinking quality.

No architectural changes to the transformer layers were required to support extended thinking; the entire innovation is in the training methodology.

## 2.3 Relation to Zen4-Pro

Zen4-Thinking shares the base model architecture with Zen4-Pro (72B dense, HHA, same layer count and dimensions). The distinction is entirely in post-training: Zen4-Thinking's post-training is optimized for reasoning quality through BRO, while Zen4-Pro's is optimized for

agentic task success through trajectory-based GRPO. The two models represent different optimization targets for the same architectural foundation.

# 3 Training

## 3.1 Pretraining

Zen4-Thinking uses the same pretraining checkpoint as Zen4-Pro (20T tokens), with a higher proportion of mathematical and scientific content (see Zen4-Pro whitepaper). The extended thinking capability is entirely post-training; the pretraining checkpoint does not include thinking token training.

## 3.2 Budgeted Reflection Optimization

BRO is a three-phase post-training procedure:

### 3.2.1 Phase 1: Cold-Start SFT

A cold-start SFT phase teaches the model the format of thinking tokens and basic extended reasoning before RL training. The cold-start corpus consists of:

- 50K mathematics problems with manually written thinking-block solutions (average 3,200 tokens per thinking block).
- 20K competition problems with expert solutions reformatted into the `<think>...</think>` format.
- 15K code problems with step-by-step algorithm development in the thinking block followed by clean implementation in the response block.

After cold-start SFT, the model can produce syntactically correct thinking blocks but has not yet learned to allocate budget proportionally to difficulty.

### 3.2.2 Phase 2: Budget-Aware GRPO

Standard GRPO [**?**] generates $G$ rollouts per problem and computes relative rewards within the group. BRO extends this with a budget-aware reward formulation.

For a problem $p$ with difficulty level $d_p$ (estimated by outcome variance at a fixed budget of 2K thinking tokens across 16 rollouts), the BRO reward is:

$$R_{\text{BRO}}(o, p) = R_{\text{outcome}}(o, p) - \lambda \cdot \max\left(0, \frac{T_{\text{think}}(o)}{B^*(p)} - 1\right) \tag{1}$$

where $T_{\text{think}}(o)$ is the number of thinking tokens in rollout $o$, $B^*(p)$ is the problem-specific budget target (a function of $d_p$), and $\lambda = 0.1$ is a penalty coefficient. This reward encourages the model to:

- Use sufficient thinking budget to solve the problem (dominated by $R_{\text{outcome}}$).
- Not exceed the problem-appropriate budget (penalized by the second term).

The budget target function $B^*(p)$ is learned from the warm-up phase data:

$$B^*(p) = \alpha_0 + \alpha_1 \cdot d_p + \alpha_2 \cdot d_p^2 \tag{2}$$

with $\alpha_0 = 512, \alpha_1 = 1024, \alpha_2 = 128$ fitted from warm-up observations.

#### 3.2.3 Phase 3: Calibration Fine-Tuning

After BRO, a final calibration SFT pass on 100K mixed-difficulty problems ensures that the model's auto-selected thinking budget matches the BRO-learned budget targets. Without this calibration, the model sometimes over-thinks at inference time despite being penalized during training.

### 3.3 Thinking-Aware Quantization

During thinking token generation, certain model layers exhibit significantly higher activation variance than during standard generation. We identify these **thinking-critical layers** by comparing activation variance statistics on thinking versus standard generation tasks across 10K samples.

In INT8 quantization, thinking-critical layers are quantized using a finer granularity (per-channel rather than per-tensor), preserving numerical precision where it matters most. This approach retains 99.1% of thinking-mode benchmark performance versus BF16, compared to 96.3% with standard uniform INT8 quantization.

## 4 Evaluation

### 4.1 Competition Mathematics

| **Benchmark** | **Zen4-Pro (standard)** | **Zen4-Thinking** | **Δ** |
|---|---|---|---|
| MATH Level 5 only | 79.4% | 93.7% | +14.3% |
| AIME 2024 (avg) | 64.3% | 87.5% | +23.2% |
| AIME 2025 (avg) | 61.8% | 84.2% | +22.4% |
| USAMO (avg '24-'25) | 48.3% | 72.4% | +24.1% |
| AMC 10 (perfect score) | 41.2% | 68.7% | +27.5% |
| OlympiadBench | 62.4% | 78.9% | +16.5% |

Table 2: Competition Mathematics: Standard vs. Extended Thinking

### 4.2 Science and Formal Reasoning

| **Benchmark** | **Zen4-Pro** | **Zen4-Thinking** |
|---|---|---|
| GPQA Diamond (0-shot) | 74.3% | 81.3% |
| GPQA Diamond (w/ thinking) | — | 84.7% |
| SciBench | 80.1% | 87.4% |
| ProofWriter (hard) | 73.4% | 88.2% |
| FOLIO (first-order logic) | 80.2% | 89.1% |
| LiveCodeBench (competition) | 72.3% | 88.2% |

Table 3: Science and Formal Reasoning Benchmarks

### 4.3 Thinking Budget Utilization

A key goal of BRO is efficient budget allocation. The following table shows average thinking token usage by problem category:

| Problem Category | Avg Thinking Tokens | Max Allowed |
|---|---|---|
| Simple arithmetic (GSM8K easy) | 128 | 32,768 |
| Math word problems (GSM8K hard) | 512 | 32,768 |
| MATH Level 3–4 | 1,840 | 32,768 |
| MATH Level 5 | 4,210 | 32,768 |
| AIME problems | 8,730 | 32,768 |
| USAMO problems | 18,400 | 32,768 |
| LiveCodeBench (hard) | 6,340 | 32,768 |

Table 4: Average Thinking Token Usage by Problem Category

The model allocates thinking budget proportionally to problem difficulty, as intended by BRO. Simple arithmetic uses <0.4% of the maximum budget; USAMO problems use 56% of the maximum. This proportionality means Zen4-Thinking is cost-efficient on easy queries while still having headroom for the hardest problems.

### 4.4 Thinking Latency

| Problem Type | Avg Latency (H100) | Avg Latency (A100) |
|---|---|---|
| Simple (128 think tokens) | 0.8s | 1.4s |
| Medium (1,840 think tokens) | 5.2s | 9.1s |
| Hard (8,730 think tokens) | 24.3s | 42.6s |
| Maximum (32,768 think tokens) | 91.2s | 159.8s |

Table 5: End-to-End Latency by Thinking Budget (including response generation)

### 4.5 Comparison Across Zen4 Family

| Benchmark | Zen4 | Zen4-Pro | Zen4-Max | Zen4-Thinking |
|---|---|---|---|---|
| MATH (Level 5) | 72.1% | 79.4% | 87.3% | 93.7% |
| AIME 2024 | 48.7% | 64.3% | 81.4% | 87.5% |
| GPQA Diamond | 67.4% | 74.3% | 79.8% | 84.7% |
| LiveCodeBench | 64.3% | 72.3% | 78.4% | 88.2% |

Table 6: Competition-Level Benchmarks Across the Zen4 Family

Zen4-Thinking provides the highest scores on competition-level tasks within the 72B parameter class, exceeding even Zen4-Max (480B/48B active) on MATH Level 5 and LiveCodeBench competition tasks. This confirms that extended thinking provides qualitatively different capability gains that cannot be replicated by parameter scaling at standard inference budgets.

## 5 Related Work

Chain-of-thought prompting [**?**] established that intermediate reasoning steps improve accuracy on multi-step tasks. Self-consistency [**?**] further improved results by sampling multiple reasoning chains and taking the majority answer. Process reward models [**?**] provided step-level feedback, enabling training directly on reasoning quality rather than final answer correctness.

Extended thinking has been explored under different framings. Scratchpad methods [**?**] demonstrated early that giving models space to work through problems helps. More recent

work on inference-time compute scaling [?] has shown that allocating more compute at inference time (through sampling, search, or extended generation) can substitute for additional training compute, representing a new axis of model capability.

BRO's budget-proportional reward extends prior work on length penalties in RLHF [?] and thinking-budget supervision from curriculum learning, applying them jointly in the GRPO setting for the first time.

# 6 Deployment

## 6.1 Thinking Mode Configuration

```
from hanzo import HanzoAI

client = HanzoAI()

# Auto-select thinking mode based on query complexity
response = client.chat.completions.create(
    model="zen4-thinking",
    messages=[
        {"role": "user", "content": "Find all integer solutions to x^3 + y^3 = z^3."}
    ],
    extra_body={
        "thinking": "auto",          # "enabled", "disabled", or "auto"
        "thinking_budget": 16384,    # max thinking tokens (default: 8192)
        "show_thinking": False       # include <think> block in response
    }
)

# Access the thinking content if show_thinking=True
if response.choices[0].message.thinking:
    print("Reasoning:", response.choices[0].message.thinking[:200], "...")
print("Answer:", response.choices[0].message.content)
```

Listing 1: Zen4-Thinking Extended Reasoning

## 6.2 Hardware Requirements

| Configuration | Hardware | Notes |
|---|---|---|
| Full BF16, 128K context | H100 × 2 | Max context |
| FP8, 128K context | H100 × 1 | Recommended |
| Q4_K_M, 32K context | A100 80GB | Budget inference |

Table 7: Zen4-Thinking Hardware Requirements

Zen4-Thinking's memory requirements during thinking generation are higher than Zen4-Pro's at equivalent context lengths, because the KV-cache grows as thinking tokens are generated. At 32K thinking tokens in BF16, the KV-cache adds approximately 12 GB beyond the base model memory requirement.

### 6.3 Recommended Use Cases

Zen4-Thinking is specifically recommended for:

- Competition mathematics and mathematical proof verification.
- Scientific hypothesis evaluation requiring multi-step chain of evidence.
- Complex algorithm design and competitive programming.
- Logic puzzles and formal verification tasks.
- Any task where the asker can verify the answer independently and cares more about correctness than latency.

Zen4-Thinking is not recommended for:

- Low-latency conversational applications (use Zen4 or Zen4-Mini).
- High-throughput summarization or extraction tasks (the thinking overhead provides no benefit on tasks that do not require multi-step reasoning).
- Interactive agentic workflows requiring fast tool-call turnaround (use Zen4-Pro).

## 7 Safety and Alignment

| Safety Metric | Standard Mode | Thinking Mode |
|---|---|---|
| Harmful content refusal | 98.2% | 97.8% |
| Over-refusal rate | 3.7% | 4.1% |
| Jailbreak resistance | 97.4% | 96.9% |
| TruthfulQA | 83.1% | 87.4% |

Table 8: Safety Metrics: Standard vs. Thinking Mode

Thinking mode shows slightly lower jailbreak resistance (97.4% vs 96.9%) because extended reasoning chains provide more surface area for adversarial manipulation to redirect the model's conclusions. Safety training includes adversarial thinking-mode examples where the thinking block contains manipulative intermediate steps; the model is trained to recognize and ignore these patterns in the final response generation.

Thinking mode shows improved TruthfulQA scores (83.1% to 87.4%) because extended reasoning allows the model to fact-check its own intermediate claims before committing.

### 7.1 Limitations

- USAMO performance (72.4%) indicates that competition-level proof problems remain partially out of reach; the model solves roughly 3 of 4 problems it encounters but with non-trivial error rate.
- Thinking token content is not guaranteed to be logically valid; the thinking block is a learned reasoning format, not a verified proof system. Users should verify mathematical claims independently.
- Long thinking chains (16K+) occasionally exhibit **reasoning drift**: the model begins exploring a correct approach but gradually transitions to a different, incorrect approach without recognizing the switch. This is flagged in the thinking block and is an active area of research.

# 8 Conclusion

Zen4-Thinking establishes that extended chain-of-thought reasoning, implemented through a principled training methodology, provides capability gains on competition-level reasoning tasks that significantly exceed what is achievable through parameter scaling at standard inference budgets. Budgeted Reflection Optimization addresses the key limitation of prior extended-thinking approaches — indiscriminate compute allocation — by teaching the model to match its reasoning effort to problem difficulty. With 87.5% on AIME 2024, 72.4% on USAMO, and 81.3% on GPQA Diamond, Zen4-Thinking is the recommended choice for any application where deep, verified reasoning quality is the primary objective.


## Acknowledgments

We thank the mathematical olympiad community whose publicly available problem sets and solutions formed the foundation of the cold-start SFT corpus, the Zoo Labs Foundation formal reasoning research team, and the Hanzo AI post-training infrastructure team.


## References

# A Model Card

| Field | Value |
|---|---|
| Model Name | Zen4-Thinking |
| Version | v2026.02 |
| Release Date | February 2026 |
| Parameters | 72.7B |
| Context Length | 131,072 tokens |
| Max Thinking Tokens | 32,768 |
| License | Apache 2.0 |
| Repository | huggingface.co/zenlm/zen4-thinking-72b |
| Documentation | papers.zenlm.org/zen4-thinking |
| Contact | research@hanzo.ai |

Table 9: Zen4-Thinking Model Card

# B BRO Ablation Study

| Training Configuration | MATH L5 | AIME 2024 | Avg Budget |
|---|---|---|---|
| Standard SFT + unconstrained GRPO | 84.7% | 71.3% | 14,200 tokens |
| + Cold-start thinking SFT | 88.4% | 78.9% | 11,800 tokens |
| + Budget-aware reward (Phase 2) | 92.1% | 85.7% | 5,340 tokens |
| + Calibration fine-tuning (Phase 3) | 93.7% | 87.5% | 4,890 tokens |

Table 10: BRO Ablation: Each Phase Contribution

The budget-aware reward (Phase 2) provides the largest benchmark gain while simultaneously reducing average thinking token usage from 11,800 to 5,340 — a 55% reduction in thinking compute. This confirms the core hypothesis of BRO: proportional allocation is more efficient

than unconstrained allocation, and the model can learn to discriminate hard from easy problems reliably.

## C Sample Thinking Token Output

The following is an abbreviated example of Zen4-Thinking's internal reasoning on AIME 2024 Problem 1 (with the thinking block truncated for space):

```
<think>
Let me read the problem carefully.
[Problem statement parsed...]

Let me try a substitution approach first.
Define x = ... [first approach developed ~1200 tokens]

That approach gives a system that doesn't close nicely. Let me try
a different angle using modular arithmetic.

Note that the equation must hold mod p for all primes p dividing ...
[modular approach developed ~800 tokens]

The constraint narrows candidates to {2, 5, 13}. Checking each:
  x=2: [check] ... valid.
  x=5: [check] ... does not satisfy the original equation.
  x=13: [check] ... valid.

So the answer is 2 + 13 = 15.
Let me verify by substituting back: [verification ~200 tokens]
Confirmed. Answer: 15.
</think>

The answer is $\boxed{15}$.
```

Listing 2: Abbreviated Thinking Block Example (AIME 2024 P1)